

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - IV

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Queue Application:

- a. Write a program to: a. Implement a queue using an array.

Code:

```
File Edit Format Run Options Window Help
from array import array

# Integer array
queue = array('i')

# Enqueue
queue.append(10)
queue.append(20)
queue.append(30)

print("Queue:", queue)

# Dequeue
print("Deleted:", queue[0])
queue.pop(0)

print("Queue after Dequeue:", queue)

print("Amit")
```

Output:

```
===== RESTART: C:/Users/mvlui/Downloads/Data
Queue: array('i', [10, 20, 30])
Deleted: 10
Queue after Dequeue: array('i', [20, 30])
Amit
|
```

b. Simulate a simple queuing system (e.g., customer service queue).

Code:

```
File Edit Format Run Options Window Help
from array import array

# Create Queue
customers = array('i')

# Add Customers
customers.append(101)
customers.append(102)
customers.append(103)

print("Customers in Queue:", customers)

# Serve First Customer
print("Served Customer:", customers[0])
customers.pop(0)

print("Queue After Serving:", customers)
```

Output:

```
===== RESTART: C:/User
Customers in Queue: array('i', [101, 102, 103])
Served Customer: 101
Queue After Serving: array('i', [102, 103])
Amit
```

Curiosity Questions:

1. Where do you see queues in daily life?
2. Why is **FIFO (First In First Out)** fair in real life? Can you think of situations where it's *not* fair?
3. What will happen if you try to **enqueue more elements than the queue size**?
4. What if you **dequeue from an empty queue**?
5. Imagine you're designing a **hospital emergency system**. Should it use a simple queue or something else? (hint: priority queue).
6. If **5 customers arrive at the same time**, but the service desk can serve only 2 at once, how would you design that?